

Garlic Mustard, a Ubiquitous Invasive Weed

Garlic mustard is an invasive weed. Learn about the characteristics that allowed it to become widespread and the steps you can take to help control this exotic weed.

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Garlic mustard (*Alliaria petiolata*) is a non-native, biennial, clumping edible weed. It is also known as sauce-alone, jack-by-the-hedge, garlic root, poor man's mustard, hedge garlic, mustard root, and garlicwort. This member of the mustard family (Brassicaceae) was introduced to North America in the 1860s by European settlers who valued it for medicinal and culinary purposes. It is now widespread throughout most of the United States. The Pennsylvania Department of Natural and

Conservation Resources (DNCR)

classifies garlic mustard as an invasive weed. As with most invasives, garlic mustard has natural predators in its native habitat, but none here, except perhaps for humans. Deer don't touch it.

All parts of the plant are edible. The roots can be used as a mild horseradish, chopped, and added to dishes. The stems and leaves have a spicy, garlicky flavor and can be chopped and used in salads, stir-fries, soups, and other cooked dishes. As with other culinary biennials like parsley, the leaves are more tender in the plant's first year and become more woody and bitter in its second year when energy turns to seed production. Its seeds can be saved to flavor food.

As a plant, garlic mustard is an opportunist. It will set seeds and quickly grow in disturbed sites. Colonies of garlic mustard are found along the edges of forests in low-light areas, especially where trees have been removed and along stream and river beds. It prefers moist sites in full or partial shade and will be more robust and fruitful under those conditions. Not known as a weed in turfgrass, home gardeners are likelier to find it in their landscape beds, along fences, and in wooded areas. It is ubiquitous as it tolerates poor growing conditions, except for highly acidic soils.

Garlic mustard is a cool-season crop. Its seeds begin germinating in March. First-year plants overwinter as green rosettes. In its native habitat, Europeans would rely on garlic mustard as a source of early-season greens. Here, this characteristic allows it to out-compete our native plants. Some of our native butterfly species lay eggs on garlic mustard. When the caterpillars emerge, they feed on this plant, but it cannot sustain them. The caterpillars either die or don't grow properly, interrupting the species' life cycle.



Garlic mustard rosette. Photo: Rob Routledge, Sault College on Bugwood.org. CC BY 3.0

In its first year, garlic mustard develops a basal rosette that will be 1 to 5 inches tall. First-year leaves are green, cordate, and rather rough. The leaf margins have rounded teeth. The leaves are variable from 1 to 6 inches long and wide. In its second year, the leaves are similar in size, though new leaves are more triangular. Second-year leaves are also dentate, but the teeth are sharply pointed. They grow alternately on quickly developing stems. The entire clump-like plant can reach from 3 to 4 feet tall. Crushed leaves smell like garlic, but the stems are not aromatic.

White flowers develop on the stem tips by May of garlic mustard's second year. The flowers grow in clusters: the inflorescence is a raceme. Individual florets have a cross-like form of 4 petals, less than an inch across. The flowers are insect-pollinated, mainly by bees, butterflies, and some moths.

As the flowers fade, seeds develop in pods called siliques. The siliques begin green, narrow, and 1.5 to 2 inches long. By June or July, the seed pods have matured and become brown and dry. The siliques will burst and spread seeds up to several feet from the parent plant. The seeds themselves are tiny (less than ¼ inch long). Each plant produces, on average, 22 siliques, each with as many as 28 seeds. Healthy mature plants can release as many as 800 seeds. The seeds are viable in the soil for up to 5 years. Thus, garlic mustard can make dense colonies. Wind, water, animals, and humans also disperse seeds. Seeds germinate all season long. By August, the second-year plant dies.

Garlic mustard has another characteristic that allows it to out-compete natives. It makes glucosinolates, an allelopathic chemical compound that inhibits the germination of other plants.

There are many ways to control this plant. In your garden, plant natives and minimize ground disturbance. Never let invasive herbaceous plants go to seed! Remove pioneering colonies when noticed. Hand removal by pulling is easy and effective. Remnants of the root left in the ground do not regrow, as do root sections of dandelions or thistles. If removing second-year plants, bag them and place them in the garbage. Do not compost. The seeds of garlic mustard can develop and become viable on stems that have been pulled (or cut) but are left lying on the soil. Reduce plant colonies by successively cutting the colony. Bag and remove the cut remnants. It may take more than one season to eliminate the colony.

Pre-emergent chemical applications are ineffective with garlic mustard as the seeds germinate over several months. Some herbicides are effective in killing garlic mustard. For specific suggestions and timing of herbicide treatments, [see the article on garlic mustard](#).

Besides controlling garlic mustard on your property, there are several actions that Pennsylvanians can take to help reduce the impact of this invasive plant in our Commonwealth. Eliminate seed transfer. Don't move firewood. Clean your footwear, clothes, bike tires, and other equipment after hiking or biking. Report sightings in our natural areas by using [iMapInvasives](#), a [citizen-science online](#) reporting and data mapping tool.

By becoming knowledgeable of the characteristics and lifecycle of specific invasive exotic plants, we all can help minimize the negative impact of these invasive weeds, including garlic mustard. Our native plants, wildlife, habitat, and fellow Pennsylvanians will all benefit from our efforts.

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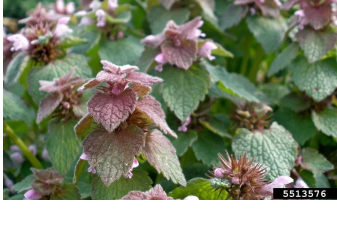
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